

Primorial-Cube Bounds and an Explicit Obstruction for a Powerful-Core Majorant of $k\sigma(k)$ Fibers

Xiaojian Zeng
PBC School of Finance, Tsinghua University, 100190
xiaojianzeng@foxmail.com

Abstract

For $f(N) = \#\{k \in \mathbb{N} : k\sigma(k) = N\}$, the question whether $f(N)$ is bounded by a fixed power of $\log N$ remains unresolved. We study only the auxiliary powerful-core majorant

$$B(N) = \#\{P : P \mid N, P \text{ is powerful, } P\sigma(P) \mid N\}.$$

First, $f(N) \leq B(N)$. For the primorial cubes $M_y = (\prod_{p \leq y} p)^3$, we prove $B(M_y) = \exp(o(\pi(y)))$. In contrast, for an explicit family N_m we prove

$$B(N_m) \geq \exp\left(\left(\frac{\log 2}{4} + o(1)\right) \frac{\log N_m}{\log \log N_m}\right).$$

Thus the filtered majorant succeeds on the natural primorial-cube stress test but cannot itself satisfy the weaker conjectural scale in general. The construction counts necessary cores, not actual preimages, and gives no lower bound for f .

1 Context and statement of results

Throughout, $\mathbb{N} = \{1, 2, \dots\}$, $\sigma(n)$ is the sum of the positive divisors of n , and all logarithms are natural. A positive integer is *powerful* if every prime dividing it occurs to exponent at least two; we include 1 as a powerful number.

The motivating problem asks whether there are absolute constants $A > 0$, $C < \infty$, and N_0 such that

$$f(N) = \#\{k \in \mathbb{N} : k\sigma(k) = N\} \leq A(\log N)^C \quad (N \geq N_0).$$

It is recorded as Problem B11 in Guy's collection [1] and as Erdős Problem 1060 [2]. This note does not resolve that problem, nor the weaker proposed estimate $f(N) = N^{o(1/\log \log N)}$.

The closest uniform bound in the sources inspected for this note is due to Kominers [3, Theorem 1.2]:

$$f(N) \leq \prod_{p \mid N} v_p(N). \tag{1}$$

The product on the right is exactly the number of powerful divisors of N . Kominers consequently obtains an upper bound with exponent constant $\log 3/3$ on the scale $\log N / \log \log N$. We retain the necessary condition $P\sigma(P) \mid N$ and analyze the smaller majorant

$$B(N) = \#\{P : P \mid N, P \text{ powerful, } P\sigma(P) \mid N\}. \tag{2}$$

Our two complementary results are as follows.

Theorem 1 (Primorial-cube bound). *Let*

$$M_y = \left(\prod_{p \leq y} p \right)^3.$$

Then, as $y \rightarrow \infty$,

$$B(M_y) = \exp(o(\pi(y))) = \exp\left(o\left(\frac{\log M_y}{\log \log M_y}\right)\right).$$

Theorem 2 (Explicit obstruction). *Let $p_1 < p_2 < \dots$ be the primes congruent to 2 (mod 3), and put*

$$N_m = \prod_{i=1}^m p_i^2 (p_i^2 + p_i + 1).$$

Then

$$B(N_m) \geq 2^m = \exp\left(\left(\frac{\log 2}{4} + o(1)\right) \frac{\log N_m}{\log \log N_m}\right).$$

The first theorem shows that condition (2) removes the primorial-cube extremals for the coarser count (1) at the target exponential scale. The second shows that this improvement is not uniform: a strategy that tries to prove the weaker conjectural bound merely by proving the same bound for $B(N)$ is obstructed. Additional information distinguishing realizable cores from the cores counted by B is indispensable. This is a bounded comparison with the sources cited above, not a claim of publication priority.

2 The powerful-core majorant

We first justify the passage from the original fiber to (2). The elementary injectivity in the next lemma is also known in greater generality: the map $n \mapsto n^a \sigma(n)^b$ is injective on squarefree integers for positive integers a, b [4, Theorem 12].

Lemma 3 (Squarefree injectivity). *The map $s \mapsto s\sigma(s)$ is injective on squarefree positive integers.*

Proof. For squarefree s ,

$$s\sigma(s) = \prod_{p|s} p(p+1).$$

Suppose two such products are equal. Cancel the factors $p(p+1)$ corresponding to primes common to the two supports. If anything remains, the two remaining prime supports are disjoint. Let r be their largest prime, and suppose it occurs on the left. On the right, r must divide q or $q+1$ for some prime $q < r$. Disjointness rules out $r = q$, so $r \mid q+1$. Since $q+1 \leq r$, this forces $q = r-1$, which is not prime when $r > 3$. If all remaining primes are at most 3, direct comparison of the four values

$$1, \quad 2\sigma(2) = 6, \quad 3\sigma(3) = 12, \quad 6\sigma(6) = 72$$

gives the same conclusion. Thus no factors remain and the supports coincide. $\square$

Lemma 4 (Core majorization). *For every $N \in \mathbb{N}$, one has $f(N) \leq B(N)$.*

Proof. Every $k \in \mathbb{N}$ has a unique decomposition $k = Ps$, where P is powerful, s is squarefree, and $(P, s) = 1$: a prime power $p^a \parallel k$ belongs to P if $a \geq 2$, and a prime with $a = 1$ belongs to s . If $k\sigma(k) = N$, multiplicativity gives

$$N = P\sigma(P)s\sigma(s).$$

In particular, $P \mid N$ and $P\sigma(P) \mid N$, so P is counted by $B(N)$. For a fixed P , the quotient $N/(P\sigma(P))$ determines at most one squarefree s by Lemma 3. Mapping each preimage k to its core P therefore proves the inequality. $\square$

There is no converse: membership in $B(N)$ does not ensure that the quotient $N/(P\sigma(P))$ is $s\sigma(s)$ for a squarefree s coprime to P .

3 Primorial cubes

Proof of Theorem 1. Every powerful divisor of M_y has the form

$$P = \prod_{p \leq y} p^{e_p}, \quad e_p \in \{0, 2, 3\}.$$

Write $S_i = \{p : e_p = i\}$. For an odd prime p ,

$$\sigma(p^3) = (p+1)(p^2+1), \quad v_2(\sigma(p^3)) = v_2(p+1) + v_2(p^2+1) \geq 2,$$

because $v_2(p^2+1) = 1$. The available 2-adic valuation in M_y is three. Thus two odd primes cannot lie in S_3 ; and if $2 \in S_3$, the factor $2^3 \mid P$ excludes every odd prime from S_3 . Hence

$$|S_3| \leq 1. \tag{3}$$

For $p \in S_2$, the corresponding factor of $\sigma(P)$ is

$$\sigma(p^2) = p^2 + p + 1 = \Phi_3(p).$$

Fix a finite set Q of primes $q \equiv 1 \pmod{3}$, and take $y \geq \max Q$. Since $v_q(M_y) = 3$, the condition $P\sigma(P) \mid M_y$ implies, for each $q \in Q$,

$$\#\{p \in S_2 : q \mid \Phi_3(p)\} \leq 3. \tag{4}$$

Apart from at most $3|Q|$ exceptions, the primes in S_2 therefore belong to the set counted by

$$A_Q(y) = \#\{p \leq y : p \text{ prime and } q \nmid \Phi_3(p) \text{ for every } q \in Q\}.$$

For $q \equiv 1 \pmod{3}$, the polynomial $\Phi_3(X)$ has exactly two roots among the $q-1$ nonzero residue classes modulo q , namely the two nontrivial cube roots of unity. The prime number theorem in arithmetic progressions for the fixed modulus $\prod_{q \in Q} q$, together with the Chinese remainder theorem, yields

$$A_Q(y) = (\delta_Q + o_Q(1))\pi(y), \quad \delta_Q = \prod_{q \in Q} \left(1 - \frac{2}{q-1}\right). \tag{5}$$

The finitely many primes dividing the modulus do not affect (5).

The sum of reciprocals of the primes $q \equiv 1 \pmod{3}$ diverges, and hence δ_Q tends to zero as Q increases through these primes. From (3)–(5), every core counted by $B(M_y)$ satisfies

$$|\text{supp } P| \leq 1 + 3|Q| + A_Q(y).$$

First let $y \rightarrow \infty$ with Q fixed and then enlarge Q . The maximum possible support size $r(y)$ is therefore $o(\pi(y))$. This order of limits uses the prime number theorem only for each fixed modulus.

Set $n = \pi(y)$. For a support of size j , each supported prime has at most the two labels 2, 3, so

$$B(M_y) \leq \sum_{j \leq r(y)} 2^j \binom{n}{j} = \exp(o(n)).$$

Indeed, $\binom{n}{j} \leq (en/j)^j$, and $r \log(2en/r) = o(n)$ whenever $r = o(n)$. Finally, the prime number theorem gives

$$\log M_y = 3\vartheta(y) \sim 3y, \quad \log \log M_y \sim \log y, \quad \pi(y) \sim \frac{y}{\log y},$$

which proves the equivalent scale in the theorem. $\square$

4 An explicit obstruction for the majorant

Proof of Theorem 2. For every subset $T \subseteq \{1, \dots, m\}$, define

$$P_T = \prod_{i \in T} p_i^2.$$

This number is powerful and

$$P_T \sigma(P_T) = \prod_{i \in T} p_i^2 (p_i^2 + p_i + 1) \mid N_m.$$

The 2^m subsets give distinct cores, proving $B(N_m) \geq 2^m$. No coprimality among the displayed blocks is required for this divisibility.

The prime number theorem in the progression $2 \pmod{3}$ gives

$$p_m \sim 2m \log m, \quad \sum_{i \leq m} \log p_i \sim m \log m.$$

Moreover,

$$\log(p_i^2 + p_i + 1) = 2 \log p_i + O(1/p_i).$$

Consequently

$$\log N_m \sim 4m \log m, \quad \log \log N_m \sim \log m.$$

It follows that

$$m \log 2 = \left(\frac{\log 2}{4} + o(1) \right) \frac{\log N_m}{\log \log N_m},$$

which is the asserted form of the lower bound. $\square$

The theorem says that $B(N)$ itself is not $N^{o(1/\log \log N)}$. It does *not* say that any of the cores P_T extends to a solution $k\sigma(k) = N_m$, much less that $f(N_m) \geq 2^m$. The original fiber may be much smaller than its majorant.

5 Exact checks and limitations

An exact depth-first enumeration represents a powerful divisor of M_y by its exponent vector in $\{0, 2, 3\}^{\pi(y)}$. For every nonzero coordinate it factors the corresponding $\sigma(p^{e_p})$, and it retains a branch precisely when no accumulated prime-adic demand exceeds the capacity three of M_y . The resulting exact counts are

y	$B(M_y)$	y	$B(M_y)$
10	4	150	16,700
20	31	200	53,771
30	35	250	120,150
50	82	300	267,096
75	1,098	400	2,676,597
100	3,323		

Direct enumeration of all $3^{\pi(y)}$ vectors independently reproduced the entries for $y = 10, 20$. The supplementary files `powerful_core_analysis.py` and `powerful_core_counts.json` contain the enumerator and its saved output. The integer counts are exact; the logarithmic ratios in the JSON file are floating-point diagnostics. Neither theorem depends on these computations.

The scope restrictions are essential. The original polylogarithmic problem and its weaker proposed bound remain unresolved here. The obstruction is to the unrestricted majorant route only, not a construction of a large fiber. The literature comparison is limited to the sources listed below, and no claim of priority is made.

Computational and AI assistance. OpenAI Codex was used for proof exploration, generation and checking of the exact enumeration code, reconstruction checks in a fresh model context, and manuscript drafting. The separate model-context check is not external peer review. The script uses exact integer arithmetic except for explicitly labelled diagnostic ratios. Human verification is recommended before public circulation. A user-supplied unpublished item on AI-assisted code generation is listed for disclosure context only [5]; it supplies no mathematical theorem, prior-art, or novelty support for this note.

References

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